Building a Chemical Language Model for RNA sequences with Masked Language Modelling

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***Abstract*—In this work, we pretrain a 4.9M parameter transformer encoder on Rfam seed sequences using masked nucleotide prediction, then evaluated its frozen embeddings with a base-pair probe trained on ViennaRNA partition-function probabilities. Using a train/test/validation split based on RNA families to prevent data leakage and pattern memorization, we compared the pretrained encoder against a random-initialization control and a rule-based canonical-pairing reference. On held-out sequences from seen families, the pretrained encoder reached AU-PRC = 0.045 compared to 0.041 for the random control (base rate 0.006). On sequences from unseen families not in the training set, the AU-PRC values were 0.052 and 0.045 (base rate 0.008). The random-initialized encoder already outperforms a canonical-pairing rule, which indicates that much of the probe's signal does not come from pretraining. The pretraining gain is modest and, on the seed-level evidence available, [TBD: stable / not stable] across seeds.**

I. Introduction

RNA function heavily depends on it's secondary structure, where the base pairs fold a sequence into stems, loops, and bulges [1]. Computational methods such as the McCaskill partition function estimate these pairings from thermodynamic parameters [2], but they are expensive for large-scale analysis. Masked language models trained on biological sequences have been shown to learn residue-level information that correlates with structure in proteins [3] and, more recently, in RNA [4]. This work asks a narrower question - does a small transformer trained only to predict masked nucleotides produce embeddings from which base-pairing probabilities can be obtained, and does that information transfer to RNA families it did not see during training?

II. Methods

A. Data

We used the Rfam seed alignments, current release Rfam 15, which contain hand-curated family alignments with family accession labels [5]. Gap characters were removed, sequences were uppercased and mapped from T to U, and only sequences of 20 to 512 nucleotides were kept. Exact duplicates were also removed. After data cleaning, 97,407 sequences from 4,192 families remained.

B. Splitting the dataset

Near-duplicate homologs in Rfam seed alignments would inflate the probe performance if they appeared in both train and test sets. We therefore clustered sequences with MMseqs2 at 80% sequence identity and 80% coverage [6], and assigned each cluster as a unit. Approximately 10% of families (421 families, 8,690 sequences) were held out entirely as a novel-family test set. All other clusters were assigned to pretraining (60%), probe training (20%), or seen-family test (20%). Table I summarizes the splits.

TABLE I
Dataset splits, sizes, and architectural roles.

Split	Sequences	Families	Role
pretrain	53,896	3,080	MLM pretraining (2% held out for validation)
probe_train	17,326	1,786	Probe training (10% held out for validation)
probe_test	17,495	1,767	Seen-family test
novel_test	8,690	421	Unseen-family test

C. Model

The encoder is a BERT-style transformer with nucleotide tokens (A, C, G, U, N) along with [PAD], [CLS], and [MASK] special tokens. It uses learned token and absolute position embeddings, pre-LayerNorm blocks, 6 layers, model width 256, 8 attention heads, feed-forward width 1024, and a maximum input length of 513 tokens including [CLS]. The transformer is built with a total of 4,874,760 trainable parameters.

D. Pretraining

The model was trained with masked language modelling. 15% of nucleotide positions were selected; of those, 80% were replaced with [MASK], 10% with a random nucleotide, and 10% left unchanged [7]. We used AdamW (learning rate = 3×10^{-4}, weight decay = 0.01), batch size = 64, linear warmup over 5% of steps, and gradient clipping at 1.0. Training ran for 20 epochs. The validation loss fell from 1.30 to 1.18 and was still decreasing at the final epoch, so the model is undertrained relative to convergence.

E. Base-pair probe

Structure targets were computed with ViennaRNA's McCaskill partition function [2], giving per-pair probabilities. The encoder was frozen, and a bilinear pair head was trained on its final hidden states: $s_{ij} = (W_q\, h_i)^T (W_k\, h_j)/\sqrt{d}$,

symmetrized. Pairs with $|i - j| < 4$ were excluded. The loss was binary cross-entropy over valid pairs against the soft ViennaRNA probabilities. The probe was trained for 30 epochs with AdamW (learning rate 10^{-3}), and the checkpoint with the lowest validation loss was retained. Evaluation used the first 500 sequences of each test set, which yields 2.4M and 1.9M valid pairs for seen and novel families, respectively.

F. Baslines

Three references were used. (1) An architecture-matched encoder with random initialization, passed through the identical probe. (2) A canonical-pairing rule that scores a pair as 1 if the two bases form an AU, GC, or GU pair and 0 otherwise. (3) The base rate of positive pairs, which is the AUPRC of a random predictor. We report AUPRC, the metric most sensitive to the rare-positive regime here, with positives defined as probability above 0.5.

III. Results

TABLE II
Base-pair probe performance (Seed 0)

Model	Seen-family AUPRC	Novel-family AUPRC
Base rate (chance)	0.0061	0.0076
Canonical-pairing rule	0.0154	0.0194
Random-init encoder	0.0410	0.0448
Pretrained encoder	0.0452	0.0522

Both encoders exceed the base rate by roughly 7-fold on both test sets. The canonical-pairing rule reaches 0.015 and 0.019, about 2.5-fold above chance and well below both encoders. The pretrained encoder improves on the random-init control by about 10% on seen families and 17% on novel families. The validation AUPRC of both probes was still rising at epoch 30, so these values may understate the converged performance of either model.

IV. Discussion

The canonical-pairing comparison is the most informative result. A bilinear head on random transformer features already recovers substantially more of the structure signal than complementarity alone. The likely source is that the frozen features encode local composition and position-dependent context, which the head can exploit even without learned sequence knowledge. The pretrained encoder adds a modest increment on top of that baseline.

The larger gain on novel families than on seen families is consistent with the pretrained model learning something that transfers beyond the families it was trained on. However, the effect is small, and a single seed cannot establish it. [TBD: state whether the ordering held across seeds.]

Absolute AUPRC values are low. Positives make up under 1% of valid pairs, and the probe was trained with an unweighted loss, so the head tends to score most pairs near zero. The metric is therefore sensitive to the head design as well as the encoder. A reader should judge these numbers against the base rate and the baselines in Table II rather than in isolation.

V. Limitations

First, the pretrained model was not trained to convergence; its MLM validation loss was still decreasing. Second, the probe is a single bilinear layer, and different heads may change the ranking of the encoders. Third, evaluation used the first 500 sequences of each test set rather than a random sample, and the probe's targets are ViennaRNA predictions, not experimental structures. Fourth, the Rfam seed alignments cover a narrow range of families compared with the full database, and sequences longer than 512 nucleotides were excluded.

VI. Conclusion

A small masked language model trained on Rfam seed sequences produces embeddings that carry base-pairing information beyond what a canonical-pairing rule captures. Its advantage over an untrained control is modest and somewhat larger on unseen families. Establishing whether that advantage is robust requires the multi-seed results, longer pretraining, and a one-hot control to separate the contribution of context from that of nucleotide identity.

References

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